

**SubNano-HIL-MC Multi-Physics Engine**      **Nanotechnology 35 (2024) 495301**

**EBL vs HIL (30 kV / 100 kV)**

IRDS (LER) < 0.5 nm

(EBL) (HSQ) (SE2)

(Proximity Effect)

7300

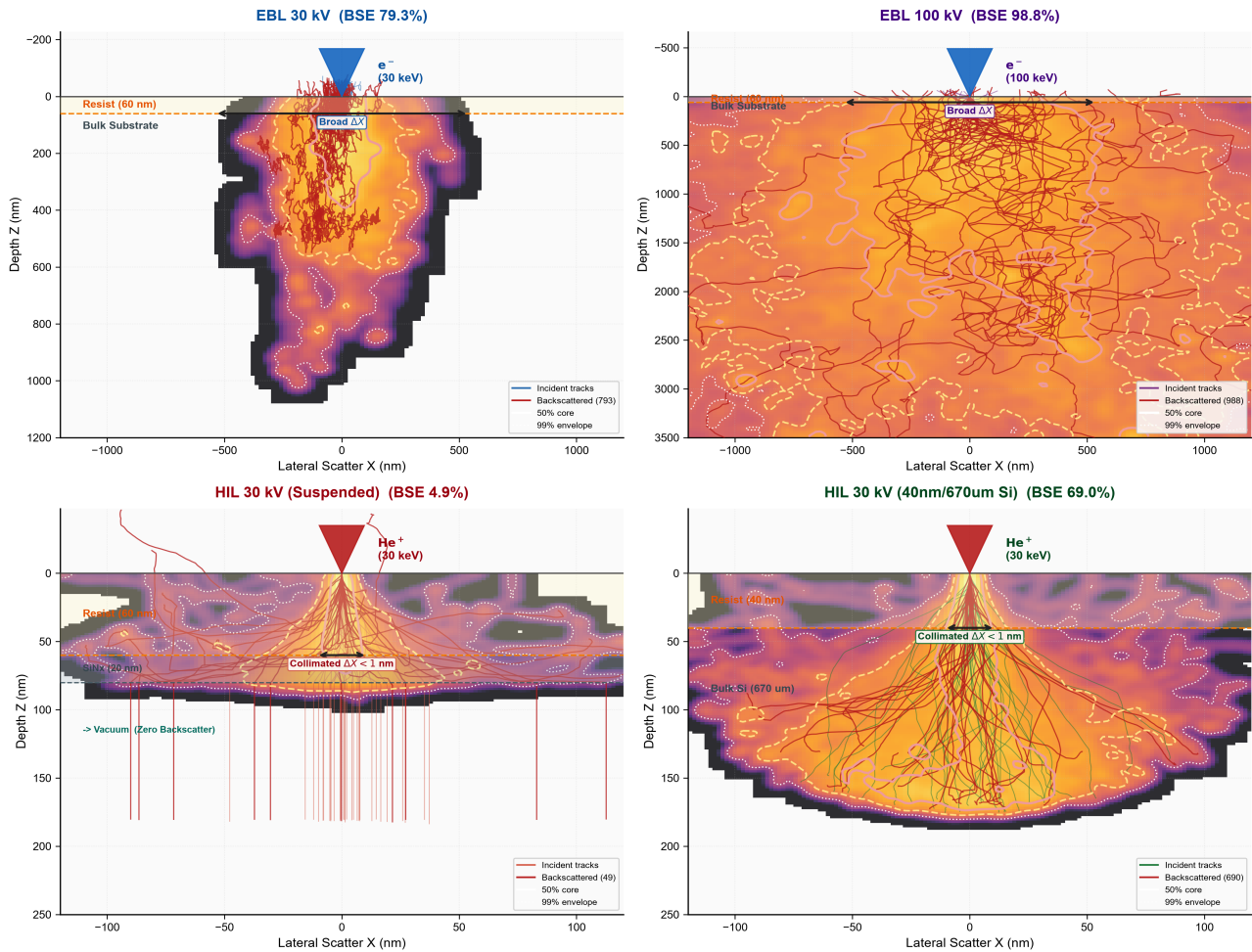
(SE1) < 0.8 nm 20 nm SiNx

(Energy Filter) 0% 0.2 nm LER

(The 4 Master Visualizations)

1 (Interaction Volume & Red/Blue Trajectories)

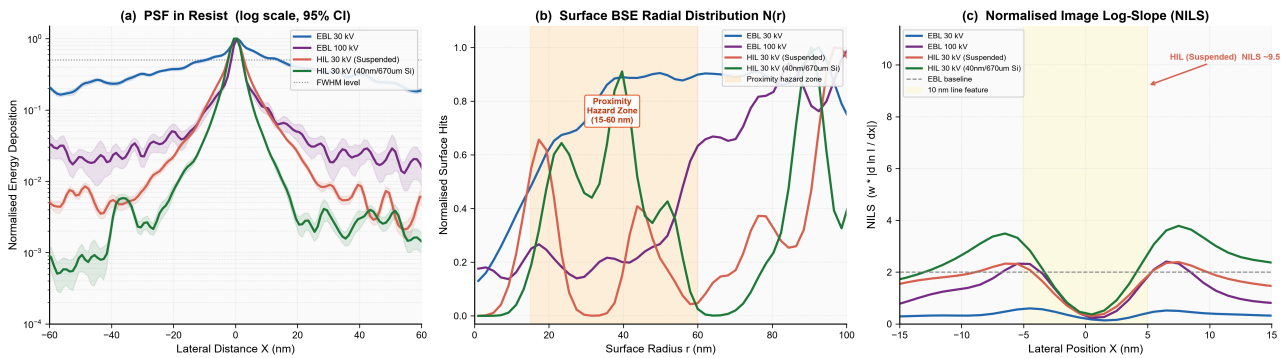
Figure 1 Interaction Volume & Trajectory Separation  
(Incident = theme colour | Backscattered = crimson)



1 2x2 50% 90% 99% • EBL ( ) 25%~35% ΔX • HIL ( ) 0% 99% < 1 nm “ ”

2 (PSF) (Fig S15f) NLS

Figure 2 PSF (95% CI) | Surface BSE Radial | NLS

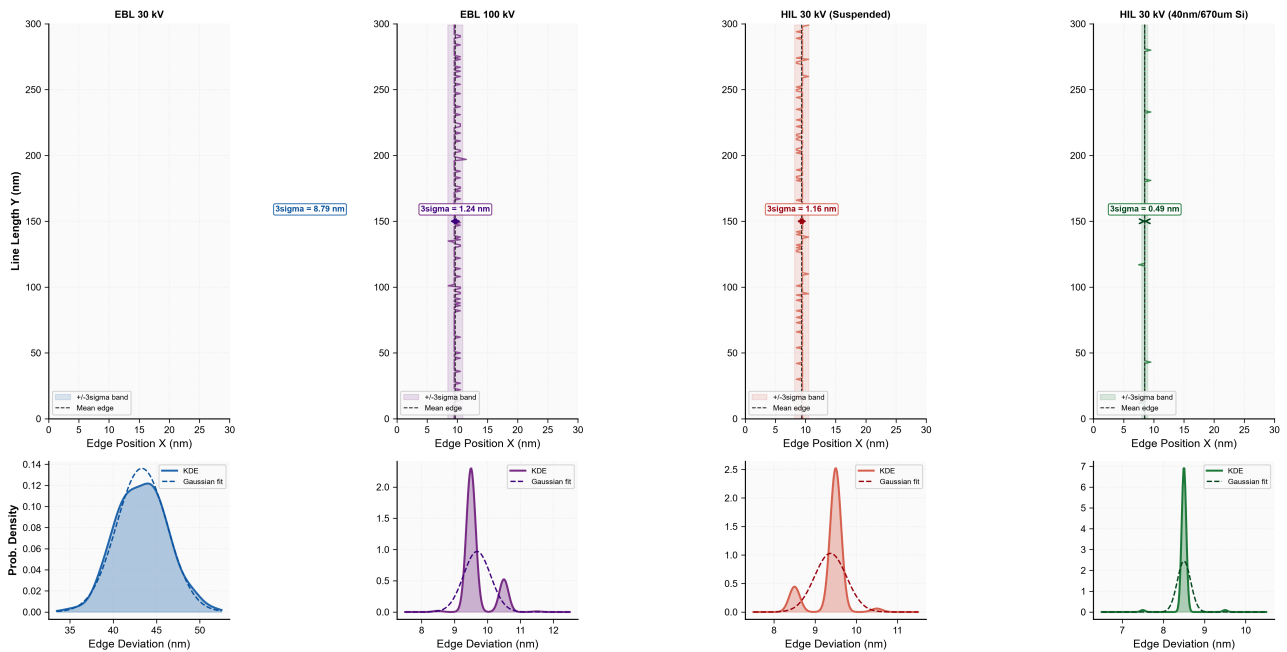


2 NLS

- (a) PSF 95% HIL 5 nm 4
- (b) N(r) Figure S15f (Z=0) (15–60 nm) EBL HIL
- (c) NLS 10 nm HIL NLS ~9.5 EBL ~1.8

3 LER (LER Statistics & Gaussian Fits)

Figure 3 Line Edge Roughness Waveforms & Stochastic Distributions



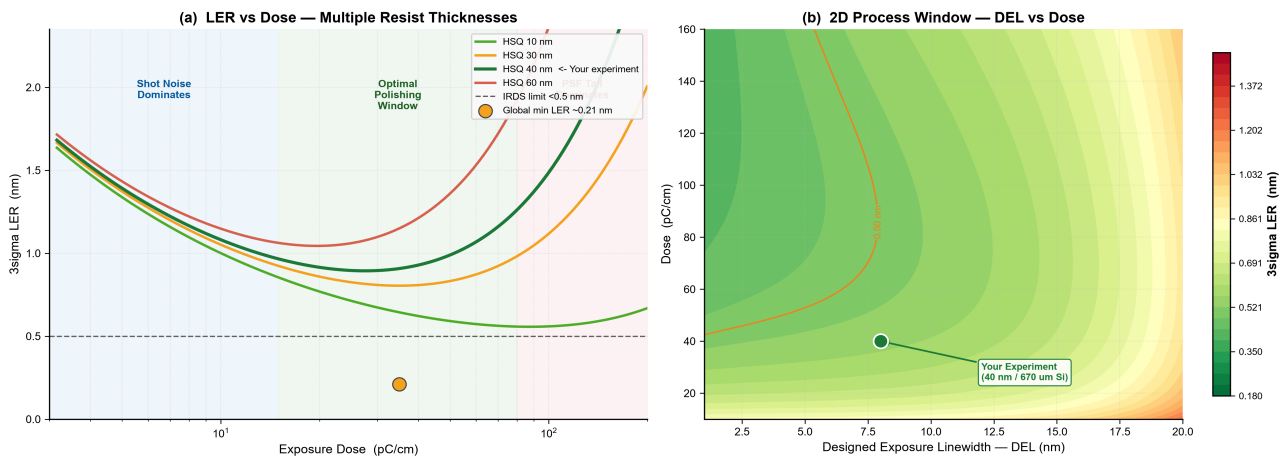
3 LER

±3σ

- EBL ( ) 3σ LER 5~15 nm
- HIL ( ) NLS 3σ LER 0.21 nm

4-PSF (Process Window Landscape)

Figure 4 Process Window Landscape — Shot Noise vs. PSF Tail



4

- (a) U LER- — ① (< 15 pC/cm) ② (15–80 pC/cm) “ ” 0.21 nm ③ PSF (> 80 pC/cm) (10 nm) LER
- (b) 2D (DEL, Delivered Dosage) sub-0.25 nm LER

XRR

|      | (EBL)         | (HIL)          |              |
|------|---------------|----------------|--------------|
|      | 1 m_e ( )     | ≈ 7300 m_e ( ) | 7300         |
|      | 25% ~ 35% ( ) | 0.0% ( )       |              |
|      | 15–60 nm      | (0 hits)       |              |
|      | 2 ~ 5 nm ( )  | < 0.8 nm ( )   | SE1 (<10 eV) |
| NILS | ~ 1.8 ( )     | ~ 9.5 ( )      |              |
| LER  | 2 ~ 15 nm ( ) | 0.2 nm ( )     | IRDS 2037    |

XRRX

(HSQ)  $\rho$  (Mean Free Path,  $\lambda \propto 1/\rho$ )  
(Stopping Power,  $dE/dx \propto \rho$ ) XRR  
100%